IDUDL: Incremental Double Unsupervised Deep Learning Model for Marine Aquaculture SAR Images Segmentation

Xinzhe Wang, Jianlin Zhou, and Jianchao Fan, *Member, IEEE*

***Abstract*— Marine aquaculture is an important natural resource exploration that requires rational planning to avoid environmental damage. Synthetic aperture radar (SAR) images are essential in remote sensing to monitor the marine ecological environment. Unsupervised methods provide an adequate solution to avoid the cost of training sample collection. However, unsupervised methods often struggle to discover effective semantic information and incrementally use newly acquired data. To address these challenges, this article presents an incremental double unsupervised deep learning (IDUDL) model, which is specially designed to characterize unlabeled marine aquaculture and achieve the results semantically. Based on the idea of alternately generating and updating pseudo-labels, the proposed IDUDL model defines the double neural networks comprising the feature extraction network (FEN) and the fully convolutional semantic segmentation network (FCSSN). A patch estimation (PE) is proposed to generate pseudo-labels with aquaculture semantic information based on the features extracted by the FEN. Then, the aquaculture extraction results are obtained by FCSSN with generated pseudo-labels. After that, the pseudo-labels and extraction results are updated in turn until the pseudo-labels are stable. In addition, due to the unique structure of double neural networks, newly acquired marine aquaculture SAR images can also be added to the pretrained FCSSN and followed pseudo-labels updated based on the FEN and PE part, which can achieve new data incremental learning without retraining the whole IDUDL model. Experiments demonstrate the effectiveness of the proposed approach based on two different ways of marine aquaculture including raft and cage types, which consist of GaoFen-3 (GF-3) and RADARSAT-2 SAR images from the Dalian and Ningde areas, respectively. The incremental experiments are also designed to verify the generalization of the IDUDL model for newly obtained marine aquaculture SAR images. The code of this work will be available at https://github.com/fjc1575/Marine-Aquaculture/tree/main/IDUDL for the sake of reproducibility.**

***Index Terms*— Incremental learning, marine aquaculture, semantic segmentation, synthetic aperture radar (SAR), unsupervised deep learning.**

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Xinzhe Wang and Jianlin Zhou are with the School of Information Science and Engineering, Dalian Polytechnic University, Dalian 116034, China (e-mail: wxzagm@dlpu.edu.cn; 20051081100492@xy.dlpu.edu.cn).

Jianchao Fan is with the Department of Ocean Remote Sensing, National Marine Environmental Monitoring Center, Dalian 116023, China, and also with the School of Control Science and Engineering, Dalian University of Technology, Dalian 116023, China (e-mail: jcfan@nmemc.org.cn).

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I. Introduction

OVER DECADES, the industry of marine aquaculture has developed rapidly [1]. The area of marine raft aquaculture in China in 2018 was 194 110 hectares, and the cage aquaculture area covered 5785 hectares [2]. Unreasonable development and utilization will bring about a series of hazards, such as harmful algal blooms and eutrophication of seawater [3], which will damage the ecological environment [4], [5], [6]. Conventional field survey is time-consuming and inefficient. Remote sensing methods have the advantages of large-area observation and timeliness and are often used in marine aquaculture extraction to reasonably monitor and plan aquaculture areas [7], [8], [9]. Most remote sensing data come from the optical remote sensing satellite, and aquaculture has prominent characteristics under optical satellites. However, natural conditions such as clouds, rain, and snow easily disturb the optical data, resulting in missing aquaculture information. Synthetic aperture radar (SAR) satellite remote sensing has the advantages of all-weather and high resolution. In addition, there are two types of marine aquaculture on China's off-coast: floating raft and cage aquaculture. They have different imaging effects on optical and SAR remote sensing images, as shown in Fig. 1. Raft aquaculture is composed of floating balls and ropes, as shown in on-site Fig. 1(a). The dense area of the floating balls under the satellite presents a long strip, as shown in Figs. 1(b) and (c). Cage aquaculture includes wood frames and nets, as on-site Fig. 1(d) presents a small rectangular box. There are a lot of speckle noises with uneven distribution in Figs. 1(c) and (f). Therefore, common machine learning methods such as the threshold segmentation will lead to poor spatial continuity.

The continuous development of deep learning has been applied to the computer vision field with great success [10], [11], [12], [13]. In recent years, the combination of remote sensing and deep neural networks has made some achievements, such as change detection, ship detection, and terrain classification, *etc*. [14], [15]. Many researchers use deep learning approaches to extract marine aquaculture information. Cui *et al.* [16] proposed an end-to-end aquaculture extraction model to overcome the "adhesion" of U-Net using the pyramid upsampling and squeeze-excitation module (PSE) structure. Shi *et al.* [17] designed a two-scale homogeneous convolutional neural network. In addition to reserving more information, sea and land segmentation and aquaculture extraction were completed. Cheng *et al.* [18] combined mixed extended convolution and U-Net to avoid wrong extraction

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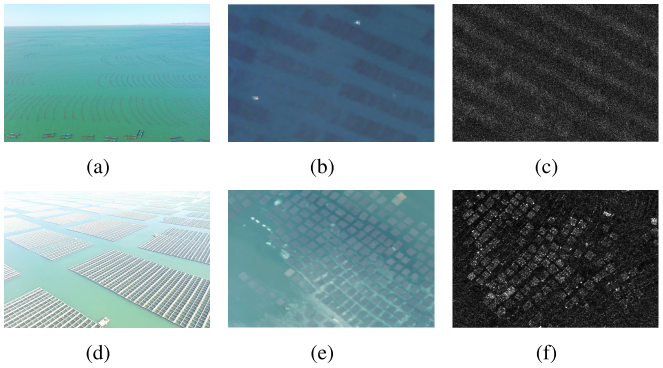


Fig. 1. Marine raft and cage aquaculture under field survey and satellites. (a) On-site image of raft aquaculture. (b) Optical image of raft aquaculture. (c) SAR image of raft aquaculture. (d) On-site image of cage aquaculture. (e) Optical image of cage aquaculture. (f) SAR image of cage aquaculture.

due to the high resolution for GaoFen-2 satellite images. The methods are mostly based on optical satellite data. For SAR satellite remote sensing, Fan *et al.* [19] considered the complex character of polarimetric SAR (PoISAR) data and proposed a complex-valued convolutional neural network that is used for marine aquaculture recognition. Wang *et al.* [20] proposed a supervised method of multiscaled attention U-Net with dilated convolution and offset convolution for SAR data. However, the above methods are based on supervised deep learning, which requires labeled samples.

Obtaining relevant labels for large amounts of remote sensing data is very difficult and time-consuming, which may severely limit the applicability and potential of supervised deep learning in practical remote sensing methods. Unsupervised learning gradually attracts attention because it does not require labeled samples. Fan *et al.* [21] proposed a novel unsupervised collective multikernel fuzzy C-means (FCM), which was applied to the remote sensing images of the Gaofen-3 (GF-3) satellite. Hu *et al.* [22] proposed a method for extracting floating raft aquaculture information from SAR images combined with a modified statistical region merging algorithm, which weakens the contamination of speckle noise. The unsupervised machine learning methods can also take full advantage of polarization information from SAR data [23], [24], [25]. Compared with the supervised methods, they solve the problem of labeled samples, but the generalization ability is weak.

The unsupervised deep learning network greatly improves the accuracy through the convolutional neural network, which can learn deep features and extract semantic information. Combining the traditional unsupervised methods with deep learning is an intuitive approach. The superpixel unsupervised algorithms are often combined with deep learning. Qin *et al.* [26] proposed a superpixel-oriented complex-valued convolutional neural network. Using superpixels instead of single pixels as the classification unit reduces the algorithm computational complexity while maintaining the image details. Kanezaki [27] and Zuo *et al.* [28] combined superpixel segmentation with feature maps extracted by the unsupervised neural networks. They used superpixel segmentation to extract image edge details. However, the number of superpixel segmentation blocks tends to be oversegmentation. In SAR images full of speckle noises, excessive focus on local information can lead to misjudgment. Most unsupervised deep learning methods are based on contrast loss functions, which explore the potential relationships between data. Generally, for similar objects, their distances are close compared with other objects. The contrastive loss designed through this idea of interclass maximization and intraclass minimization is generally used in the unsupervised deep learning networks [29], [30], [31]. Recently, researchers have tried to apply unsupervised deep learning to remote sensing images. Kang *et al.* [32] used the first law of geography, which means that regions with closer distances are more similar than regions with farther distances, to construct a contrastive learning loss and use momentum updates. The region is obtained according to the outward expansion of the central pixel, which determines that this region is a regular quadrilateral, which is less suitable for image segmentation tasks of irregular objects. Liu *et al.* [33] used the rich band information of hyperspectral images to construct the contrastive loss, which is formed by imaging in different bands at the same location. Tulczyjew *et al.* [34] proposed an unsupervised method of the recurrent neural network, which takes the hyperspectral band dimension as a sequence, trains the recurrent neural network to find the essential features of the band through the encoding–decoding reconstruction, and completes the segmentation by the clustering method. Saha *et al.* [35] used multitemporal image changes to assign meaningful pseudo-label clusters according to the consistency of changes to complete high-resolution remote sensing image segmentation. Obviously, the former uses rich band information, and the latter uses rich multitemporal high-resolution image information to improve the extraction accuracy of unsupervised deep learning through more information. However, both hyperspectral data and long-term high-resolution data are difficult to obtain, and the data requirements are relatively high, so it is difficult to apply to a single SAR image.

In practical application, with the formation of satellite networks, there are more and more satellite data. Faced with a large number of unlabeled samples, unsupervised deep learning poses new challenges, requiring incremental approaches [36]. Based on the trained unsupervised networks, the new data are processed on time, and the results after retraining of the trained network are effectively improved, which is called an incremental network. When new data come into the model, assuming the assigned pseudo-labels as ground truth, the outputting network is retrained. This idea, also known as self-labeling [37], becomes relevant to simply annotating unlabeled data since this added knowledge allows the neural network to refine its behavior to correctly adapt to more complex samples of the target set. This incremental process is repeated until the entire target set is thoroughly annotated and the assigned pseudo-labels are stable.

Because of the above problems, this article proposes an incremental double unsupervised deep learning (IDUDL) model. It is specially designed for extracting marine aquaculture information from SAR data, solving semantic learning for different imaging features, incremental utilization of a large number of unlabeled samples, and unsupervised deep learning. It has double networks, including the feature extraction network (FEN) for assigning the pseudo-labels and the fully convolutional semantic segmentation network (FCSSN) for getting the output. The semantic segmentation network is inspired by the "adhesion" phenomenon exhibited by the

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U-Net model in marine aquaculture extraction. After the feature extraction network and superpixel segmentation, patch estimation (PE) is used to ensure that the pseudo-label contains the edge semantic information of aquaculture. Through the semantic pseudo-labels and the "adhesion" characters, the FEN is guided to optimize the pseudo-labels while the segmentation results improve from the FCSSN. Finally, the purpose of alternately updating the FEN and the FCSSN is achieved. This article makes the following contributions.

1) A double unsupervised deep learning model is proposed for marine aquaculture, which consists of the FEN and the FCSSN. The FEN extracts the deep feature of aquaculture for generating the pseudo-labels. The FCSSN obtains the results of aquaculture for updating the pseudo-labels. The FEN and FCSSN are updated alternately to achieve the alternately generating and updating pseudo-labels.
2) A PE is proposed to find aquaculture semantic information, which is converted from an irregular feature distribution into a spatially contiguous patch. The PE constrains the generated pseudo-labels to conform to aquaculture semantics, achieving the deep neural network explainability.
3) For new data, the proposed model shows incremental learning in alternating processes. The new data generate pseudo-labels via FEN based on the results of the pretrained FCSSN. In this way, the IDUDL model is retrained on new data and quickly reaches a state of convergence. Incremental training is effective in improving the accuracy of the extracted results.

The remainder of this article is organized as follows. Section II introduces the related work related to the research direction of this article and provides prior knowledge. Section III introduces the proposed IDUDL model in detail for SAR image aquaculture. Section IV shows the experimental results of the work. Finally, Section V summarizes the highlights of the article and future research.

II. Preliminary Work

A. Unsupervised Deep Learning

Mainstream unsupervised deep learning is divided into two categories, one is based on contrastive learning, and the other is based on unsupervised constraints. Many works are carried out based on contrastive loss on contrastive learning. The core idea is feature invariance. The feature similarity degree of the same feature in different views is much higher than that of different features. Such ideas are also applied to mutual information loss. Ji *et al.* [38] proposed invariant information clustering (IIC), and the loss function is designed according to the information theory. The network can directly output semantic labels by maximizing the mutual information assigned by the same object clusters of the two views. Spatial information between image patches can be obtained, and image segmentation is derived by image patches' clustering. The information theory method is used for image segmentation, grasping the changes in adjacent pixels in the same image, and combining the clustering method based on mutual information with the deep network.

The other type combines unsupervised and deep learning, expecting to generate better pseudo-labels, mostly using methods such as clustering or superpixel segmentation and optimizing pseudo-labels. Kanezaki [27] proposed a superpixel deep learning (SDL) method. The unsupervised image segmentation method sets three criteria for image segmentation, similar pixels with the same label, spatially continuous pixels with the same label, and the number of unique labels is desired to be large. Design the loss function according to the criteria and enhance its results using the superpixel method. Kim *et al.* [39] tried to use the $L1$-norm instead of superpixel segmentation to get spatial continuity and proposed a differentiable feature clustering network (DFCNN). The DFCNN obtains pseudo-labels C_i from the network output

$$C_i = \left\{ r'_n \in \mathbb{R}_q | r'_{n,i} \geq r'_{n,j}, \forall j \right\} \tag{1}$$

where r'_n is the network outputs, q is the number of output channels, n represents the pixel, and the i, j is the value of output channels. This is an intuitive strategy for obtaining pseudo-labels and is also used in FEN of the IDUDL model.

The DFCNN designed total loss L, including similarity loss L_{sim} and continuous loss L_{con}. L_{sim} is the cross-entropy loss of pseudo-labels and network output, which can make the unique label as few as possible. The network optimization direction is constrained by L_{con} to prevent the output from tending to one class

$$L = L_{\text{sim}}\left(\left\{r'_n, c_n\right\}\right) + \mu L_{\text{con}}\left(\left\{r'_n\right\}\right) \tag{2}$$

$$L_{\text{sim}}\left(\left\{r'_n, c_n\right\}\right) = \sum_{n=1}^{N}\sum_{i=1}^{q} -\delta(i - c_n)\ln r'_{n,i} \tag{3}$$

where

$$\delta(t) = \begin{cases} 1, & \text{if } t = 0 \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

L_{con} is composed of the $L1$-norm. Based on the idea of space invariance, the image that shifted to the left by one bit and the original image should be the same. The same is true of moving up

$$L_{\text{con}}\left(\left\{r'_n\right\}\right) = \sum_{\xi=1}^{W-1}\sum_{\eta=1}^{H-1} \left\| r'_{\xi+1,\eta} - r'_{\xi,\eta} \right\|_1 + \left\| r'_{\xi,\eta+1} - r'_{\xi,\eta} \right\|_1 \tag{5}$$

where ξ represents the width of the image, and η represents the height of the image.

The network constructs a classifier that is the strategy of assigning channels to generate pseudo-labels and complete the unsupervised segmentation task.

B. SuperPixel Segmentation

Superpixel segmentation refers to irregular pixel blocks with specific visual significance composed of adjacent pixels with similar texture, color, brightness, and other characteristics. The pixels are grouped by the feature similarity so that a small number of superpixel patches replace a large number of pixels to express image features, which greatly reduces the complexity of image postprocessing and is usually used as a preprocessing step in segmentation algorithms.

Simple linear iterative clustering (SLIC) [40] has a high comprehensive evaluation in computing speed, object contour

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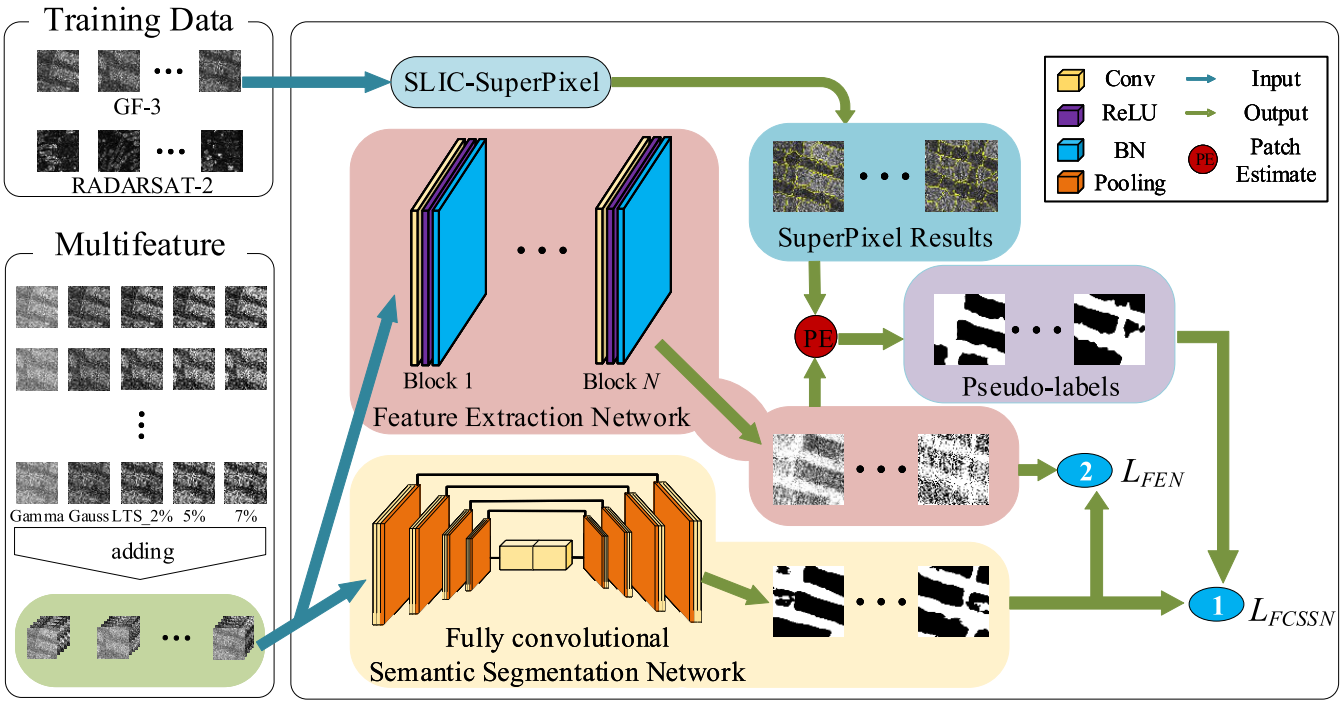


Fig. 2. Structure of the IDUDL model, which includes: image enhancement for multifeature adding, pseudo-labels from PE between the FEN and the SLIC-superpixel algorithm, and the output results from the semantic segmentation network. L_{FCSSN} is the loss function of FCSSN. L_{FEN} is the loss of FEN.

retention, and superpixel shape. The SLIC algorithm converts the original image into the Commission Internationale de L'Eclairage Lab (CIELAB) color space. The SLIC adapts the K-means clustering to generate superpixels, which considers the color similarity and spatial similarity to calculate the Euclidean distance

$$D = \sqrt{d_c^2 + \left(\frac{d_s}{S}\right)^2 m^2} \tag{6}$$

where $d_c = ((l_j - l_i)^2 + (a_j - a_i)^2 + (b_j - b_i)^2)^{1/2}$, l, a, and b denote the values of color channels L, A, and B, respectively, and d_c is the CIELAB distance of pixels and clustering center. $d_s = ((x_j - x_i)^2 + (y_j - y_i)^2)^{1/2}$, x and y denote the horizontal and vertical coordinates of pixels, respectively, and d_s is the spatial distance of pixels and clustering center. $S = (N/k)^{1/2}$, and it is determined by the image size N and the initial number of clustering centers k. m is a constant that determines the maximum color distance.

After each pixel point is assigned to the nearest clustering center, the average is taken in channels L, A, B, X, and Y of pixels under the same clustering center to update the clustering center. According to the L_2-norm, the new and previous residual errors of clustering center location are calculated. Allocate pixels and update clustering centers iteratively until the error converges.

The SLIC narrows the search domain and gets faster, more memory-efficient, and more accurate results. Commonly used in the preprocessing of image segmentation tasks, an image can be divided into many small blocks with certain morphological features of different sizes and shapes, which provide semantic information in the case of entirely unlabeled.

III. Proposed Method

To build an unsupervised deep learning network for aquaculture, the idea of this proposed method is to generate pseudo-labels and achieve optimization of pseudo-labels and outputs alternately. The proposed method can be summed up in four parts, and the framework is shown in Fig. 2. The first part is the multifeature addition of the aquaculture training set. The second part is the PE between the SLIC-superpixel and the FEN to generate pseudo-labels. The third part is the semantic segmentation network training with the pseudo-labels to get extraction results of aquaculture. The fourth part is to feed the results back to the pseudo-labels. Eventually, FCSSN and FEN are updated alternately. The FEN extracts aquaculture deep features using a simple network structure. It can be composed of any module, and each module includes a convolution layer, an activation layer, and a normalization layer. The semantic segmentation network is used to obtain aquaculture segmentation results. It can be composed of any semantic segmentation network.

A. IDUDL Model

The actual problem solved by this network is described below. In detail, let $\mathcal{X} = \{x_1, \ldots, x_M\}$ be an SAR dataset composed of M images. Considering the enhanced contrast between the aquaculture area and the sea area, image enhancement is performed for each original image x_i. Each type of operation has special effects. The linear truncation stretching 2%, 5%, and 7% are used to highlight the aquaculture area while avoiding outliers. The Gamma transform stretches the areas with higher gray levels in the image while compressing the parts with lower gray levels, which makes the sea area darker. The Gaussian filtering smooths the aquaculture area

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continuously. Then, a five-channel enhanced image $x_{i_a} \in \mathbb{R}^5$ is generated. The SLIC algorithm obtains the superpixel segmentation result s_i of each image x_i, and each image is divided into n patches to represent as s_i^n. The obtained superpixel segmentation results conform to the aquaculture semantics and can provide prior knowledge for the generation of pseudo-labels. x_{i_a} is input into the FEN $f(x_{i_a}; \theta_f)$ to obtain the deep feature map r_{i_a} by indexing the maximum value between the output channels of the FEN. Based on the superpixel segmentation map $s_i^j, j \in \{1, \ldots, n\}$, the same area patch $r_{i_a}^j, j \in \{1, \ldots, n\}$ in the deep feature map is performed the PE, which is described in detail in Section III-B, to generate the pseudo-labels $c_i, i \in \{1, \ldots, M\}$.

With pseudo-labels, the enhancement images x_{i_a} are input into the FCSSN $g(x_{i_a}; \theta_g)$. It takes the pseudo-labels c_i as the target. To avoid falling into local optimization, appropriate number of iterations τ should be selected to get the aquaculture results x_g. For updating the pseudo-labels, the result x_g is used as the target of FEN. The characteristics of the semantic segmentation network can be used to achieve dynamic optimization of pseudo-labels that are close to real labels. In summary, the IDUDL fixed the pseudo-labels to train the semantic segmentation network and get the aquaculture results. Then, the IDUDL fixed the results to train the FEN and get the new pseudo-labels. Finally, the new pseudo-labels are used to train the semantic segmentation network again to generate new results to achieve pseudo-labels and the aquaculture results updated alternately.

Due to fully unsupervised learning, in the first round of training, the FEN does not come from the results of the semantic segmentation network as the target. So the FEN in the first round focuses more on the image itself features about pixel similarity and spatial continuity in the image. The first round of FEN targets is the result of PE. For example, the feature map is punctiform because of the speckle noise in the SAR image. Then, the PE gets spatial continuity pseudo-labels as the network targets in the first round. However, it will create a problem that the FEN can easily fall into overfitting. Since the network's target is itself, much aquaculture information will be lost, and the aquaculture area will be missed. So, the first is to reduce the number of training iterations κ, and the second is to use the PE results instead of directly using the feature network output. Also, the number of training iterations τ of the semantic segmentation network is worthy of attention. Too large or small can easily lead to problems such as falling into the local optimum and failing to converge. How to set the hyperparameters κ and τ will be discussed in Section IV-F.

The iterative termination condition in the IDUDL model is the difference between the labels for each round of training. In unsupervised learning, there is no ground truth to judge the quality of learning. It is necessary to find a stable state of pseudo-labels and set the iteration termination condition ε by comparing the difference δ of each round of the pseudo-labels.

B. Patch Estimation

Before the pseudo-labels are generated, the output of the FEN has two channels corresponding to the aquaculture area and sea in the marine aquaculture extraction task, finding the maximum value between the two channels and the output of the channel index. Then the two channels convert into the one-channel deep feature map with two classifications r_{i_a}. The SLIC-superpixel results s_i are obtained from the original image. As far as the superpixel algorithm is concerned, a whole image is divided into multiple patches s_i^j. The location of patches can be mapped onto the feature map $r_{i_a}^j$.

The PE is the dominance ratio judgment for a single patch. Since the feature map contains two indexes of the channel, it can be judged whether the index of the first channel accounts for more or the second channel accounts for more in a single patch and assign more to the entire patch. For example, the index value 0 for aquaculture is more than another index value 1 for others in the first patch of the first feature map $r_{1_a}^1$, and then all the values in the $r_{1_a}^1$ position are assigned as the aquaculture, and so on, and each patch of each image is unified. Finally, the pseudo-labels c_i from $r_{i_a}^j, i \in \{1, \ldots, M\}, j \in \{1, \ldots, n\}$ after PE are obtained.

Using this PE method, the pseudo-labels can get aquaculture edge information and greatly enhance the spatial continuity. In the first round of training discussed above, the pseudo-labels generated from FEN easily caused overfitting. The PE method adds a weight decrement strategy. The dominance ratio judgment of PE decreases gradually in the first round of training. For example, the initial threshold for dominance ratio judgment is 0.5, which gradually decreases as the FEN is trained iteratively. This can improve the problem of the spatial loss caused by FEN overfocusing on pixel similarity in unsupervised situations.

C. Loss Function

In the IDUDL model, the target loss function consists of three components, namely, L_{FEN_0}, L_{FEN_n}, and L_{FCSSN}. L_{FEN_0} is the FEN loss in the first round, and L_{FEN_n} is the FEN loss in $n > 1$ round. L_{FCSSN} is the semantic segmentation network loss. In detail, the deep feature and the patch position output by the superpixel algorithm are used for PE. The superpixel patches are mapped to the deep feature map. The dominant feature point value will replace the entire superpixel patch to generate pseudo-labels. After inputting multiple features into the FEN, the network is updated through the pseudo-labels generated by PE. The loss function is as follows:

$$L_{\mathrm{FEN}_0} = \sum_{n=1}^{N} \{\mathrm{PE}(f(x_n), \mathrm{SLIC}(x_n)) \log[f(x_n)] + [1 - \mathrm{PE}(f(x_n), \mathrm{SLIC}(x_n))] \log[1 - f(x_n)]\} \quad (7)$$

$$L_{\mathrm{FEN}_n} = \sum_{n=1}^{N} \{\mathrm{PE}(f(x_n), \mathrm{SLIC}(x_n)) \log[g(x_n)] + [1 - \mathrm{PE}(f(x_n), \mathrm{SLIC}(x_n))] \log[1 - g(x_n)]\} \quad (8)$$

where x is the input image, N is the number of input images, $\mathrm{SLIC}(\cdot)$ is the superpixel segmentation function, PE is the PE method, $f(\cdot)$ is the FEN model, and $g(\cdot)$ is the FCSSN model.

As seen above, the difference between the first round of loss and others is the target in the FEN. In the first round of training, the target is the PE results, and the others are the semantic segmentation network outputs.

The outputs of the semantic segmentation network are optimized to extract the aquaculture areas, and the loss function

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is as follows:

$$L_{\mathrm{FCSSN}} = \sum_{n=1}^{N} \{g(x_n) \log[f(x_n)] + [1 - g(x_n)] \log[1 - f(x_n)]\} \tag{9}$$

The outputs of the semantic segmentation network are used to guide the FEN to optimize the pseudo-labels. This requires that when choosing the semantic segmentation network, its results are better than the pseudo-labels. For a specific problem, it is necessary to comprehensively analyze the network and understand its characteristics. In this way, the output generation depends on both the pseudo-labels and affects the pseudo-labels in reverse. The proposed method realizes the gradual progress of alternating updates.

Algorithm 1 IDUDL Model

Require: $x_i, \tau, \kappa, \varepsilon$
$x_{i_a} \leftarrow ImageEnhancement(\{x_i\})$
$s_i^j \leftarrow SLIC(x_i)$
$x_i^j \leftarrow GetPatch(\{s_i^j\})$

for $True$ **do**
  Initialize: θ_f
  for $t = 0$ **to** κ **do**
    $r_{i_a}^j \leftarrow GetFeature(\{f(x_{i_a}^j; \theta_f)\})$
    for $j = 0$ **to** n **do**
      $c_i \leftarrow PatchEstimation(f_{i_a}^j)$
    end for
    $L_0 \leftarrow L_{FEN}(\{r_{i_a}, c_i\})$
    $L_t \leftarrow L_{FEN}(\{r_{i_a}, Output_{FCSSN}\})$ // $t > 0$
  end for
  Out c_i

  Initialize: θ_g
  for $e = 0$ **to** τ **do**
    $Output_{FCSSN} \leftarrow Segmentation(\{g(x_{i_a}; \theta_g)\})$
    $L \leftarrow L_{FCSSN}(\{Output_{FCSSN}, c_i\})$
  end for
  Out $Output_{FCSSN}$
  if $\delta < \varepsilon$ **do**
    break
end for

D. Incremental Network

Marine aquaculture monitoring is a long-term task with a large amount of new data every day. To avoid repeated training of new aquaculture data, it is expected to use the existing aquaculture extraction network to obtain prior knowledge to save training costs. In the IDUDL model, there is an incremental training advantage due to its double-network structure. Typically, the final aquaculture results are obtained through the semantic segmentation network. When the IDUDL model converges, there is a trained semantic segmentation network for aquaculture extraction. The accuracy of the results obtained directly through the semantic segmentation network decreases when new aquaculture data are obtained. However, the semantic segmentation network output also has a reference value, similar to pretraining. Optimizing the IDUDL model to achieve higher accuracy and better generalization is the goal of implementing the incremental network.

After new data are entered, the problem is that new data require new pseudo-labels. Due to the generalization problem, the semantic segmentation network's aquaculture extraction results of new data cannot be directly used as pseudo-labels. However, it can be used as prior knowledge to input the FEN. After constraining the aquaculture semantic information through PE, it can be used as a pseudo-label to start new training of the IDUDL model.

In unsupervised learning, it is necessary to learn a large amount of new data. The more the data, the more potential rules can be found, and higher accuracy can be achieved. The IDUDL model achieves this through its double-network feature, which can both output results and generate pseudo-labels.

IV. Experiments

The proposed IDUDL model has two parts of network parameters that need to be discussed, namely, the FEN and the fully convolution semantic segmentation network. In the FEN, the convolutional kernels are 3×3. To prevent the output from being inconsistent with the input size, the network adds a padding operation without adding a pooling layer. The FEN is only composed of convolution, which assists the batch normalization and rectified linear unit (ReLU) to build $N = 5$ layers of convolutional neural network. The input channel is five after image enhancement, the number of intermediate channels is set to 100, and finally, the feature map of two channels is output to facilitate the formation of pseudo-labels with the maximum channel index. The fully convolution semantic segmentation network is built concerning the U-Net network, which consists of 3×3 convolutional kernels, 2×2 max-pooling size, and 64, 128, 256, and 1024 number of channels.

For the SLIC superpixel algorithm, the number of segmentation patches is 30 for the GF-3 raft aquaculture data and 120 for the RADARSAT-2 cage aquaculture data. The search domain is five for the GF-3 raft aquaculture data and three for the RADARSAT-2 cage aquaculture data.

All the experiments are implemented in PyTorch1.7, with an Intel Xeon Silver 4208 with 2.1 GHz and an Nvidia GeForce RTX 3080.

A. Evaluation Criteria

The unsupervised methods need to find the latent properties of the data to define classification results. According to the imaging characteristics of the aquaculture area in the SAR image, the surface roughness results in larger backscattering coefficients in aquaculture than on the sea surface. To keep the unsupervised results consistent, they are reassigned classes based on the corresponding pixel values of the original image. Considering target class pixels as positive and all other pixels as negative, multiple evaluation indicators are introduced, including mean intersection over union (MIoU), Kappa coefficient, overall accuracy (OA), precision (P), and recall (R). The formula is as follows:

$$\mathrm{OA} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \tag{10}$$

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$$R = \frac{T_P}{T_P + F_N} \tag{11}$$

$$P = \frac{T_P}{T_P + F_P} \tag{12}$$

$$\text{MIoU} = \frac{1}{2}\left(\frac{T_P}{T_P + F_P + F_N} + \frac{T_N}{T_N + F_N + F_P}\right). \tag{13}$$

where T_P is the number of pixels correctly predicted as the aquaculture area, T_N is the number of pixels correctly predicted as the sea area, F_P is the number of pixels that predict sea as aquaculture, and F_N is the number of pixels that predict aquaculture targets as the sea

$$\text{Kappa} = \frac{\text{OA} - P_e}{1 - P_e} \tag{14}$$

where

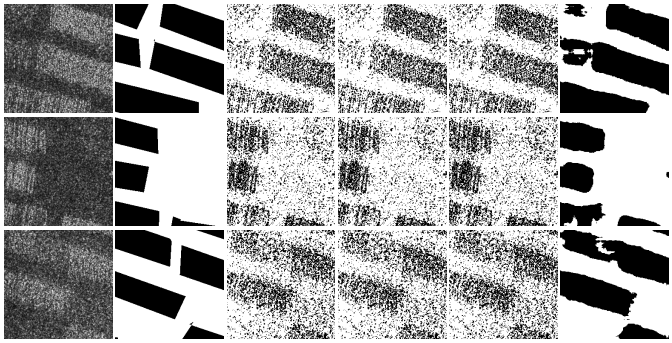
$$P_e = \frac{(T_N + F_P) \times (T_N + F_N) + (T_P + F_P) \times (T_P + F_N)}{N \times N} \tag{15}$$

The aquaculture area extraction is not only an image segmentation task but also a pixel-level classification task. The MIoU calculates the average of the ratios of the intersections and unions of all the categories for the image segmentation tasks, measuring the spatial coincidence between the prediction and the ground truth. Kappa is a measure of classification accuracy. Usually, Kappa falls between 0 and 1. It can be divided into five groups to represent different levels of consistency: 0.0–0.20 for slight consistency, 0.21–0.40 for fair consistency, 0.41–0.60 for moderate consistency, 0.61–0.80 for substantial consistency, and 0.81–1 for almost perfect consistency. The accuracy is the most common and basic evaluation. However, in the case of binary classification and unbalanced positive and negative samples, especially when a small number of sample classes is interested, accuracy is barely satisfactory. Therefore, precision and recall are introduced to evaluate the model further. If the model is greedy and wants to cover more samples, it has the potential to make mistakes. At this time, the recall value is often high, and the precision is low. If the model is conservative and only makes predictions for certain samples, the precision value is often high and the recall value is low.

B. Dataset

There are mainly two ways of marine aquaculture, the raft and cage aquaculture, as shown in Fig. 1. The raft aquaculture referred to the use of floats and ropes to form floating rafts on the surface of shallow water and fixed to the seabed with cables. The seaweed and shellfish seedlings are suspended on the sling. Imaging in SAR is mainly affected by the floating ball [41]. Near-shore cage aquaculture is primarily composed of a frame system, box system, and mooring system, among which the frame system is assembled chiefly from plates, buoys, and other components. Imaging in SAR is mainly affected by the frame system. The common point of imaging is that they are both artificial buildings. Under horizontal transmit and horizontal receive (HH) polarization, there is a clear difference compared with the sea surface.

To demonstrate the effectiveness of the proposed method, the SAR image datasets were selected from different satellites, different regions, and different aquaculture types. The raft aquaculture dataset is from the GF-3 satellite. The resolution is 3 m in Changhai County, Dalian, Liaoning Province. The cage aquaculture dataset is from the RADARSAR-2 satellite. The resolution is 8 m in Sanduao County, Ningde, Fujian Province.



(a) (b) (c) (d) (e) (f)

Fig. 3. Experimental results of raft aquaculture based on the classic unsupervised learning comparison methods. (a) Original image. (b) Ground truth. (c) OTSU. (d) K-means. (e) FCM. (f) IDUDL model.

The GF-3 raft aquaculture dataset is cropped to size 256×256, and a total of 164 pictures are used as the test pictures, which were used to test all the comparative methods, both supervised and unsupervised. The supervised network selects another 208 images for training and 89 images for verification, for a total of 297 images. At the same time, these 297 images are also used as part of the incremental experimental.

The RADARSAR-2 cage aquaculture dataset is also cropped to size 256×256, and a total of 121 pictures are used as the test pictures, which were used to test all the comparative methods, both supervised and unsupervised. The supervised network selects another 61 images for training and 26 images for verification, for a total of 87 images. Although the resolution is different, the image crop size is not changed. The experimental setup is kept as consistent as possible to demonstrate the effectiveness of the method.

The ground truths are acquired from cruising measurement with differential global positioning system (GPS) in Changhai survey areas. Some representative samples (control points), such as inflection points and boundary points, are obtained from the level of on-site investigation.

C. Raft Aquaculture

In this article, the classical unsupervised methods, K-means, OTSU, and FCM, unsupervised deep learning methods, IIC, DFCNN, and SDL, and supervised deep learning methods, U-Net and SegNet, are compared with the proposed IDUDL model. The experimental results of the raft aquaculture by GF-3 are shown in Figs. 3–5, respectively. Moreover, the corresponding accuracy is tabulated in Table I. The best results are marked in bold in the tables.

The classical unsupervised experiments indicate that the IDUDL model can avoid the interference of speckle noise in the seawater area. From Fig. 3(a), the raft aquaculture is a floating ball with distance on the sea surface, and the problem of speckle noise is particularly prominent. The classical unsupervised methods are judged based on pixel values, which are susceptible to speckle noise and generate many misjudgment points. The OA seems good due to the apparent difference

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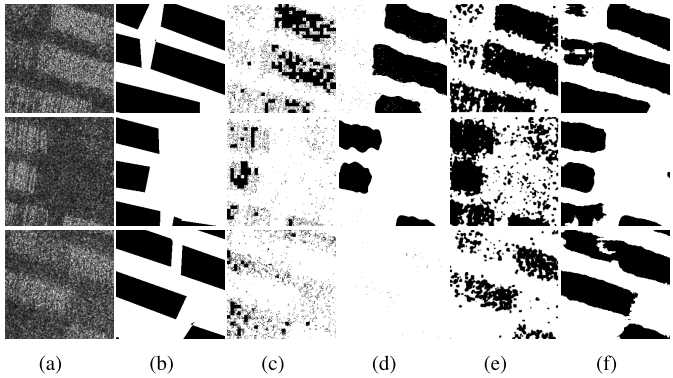


Fig. 4. Experimental results of raft aquaculture based on the unsupervised deep learning comparison methods. (a) Original image. (b) Ground truth. (c) IIC. (d) SDL. (e) DFCNN. (f) IDUDL model.

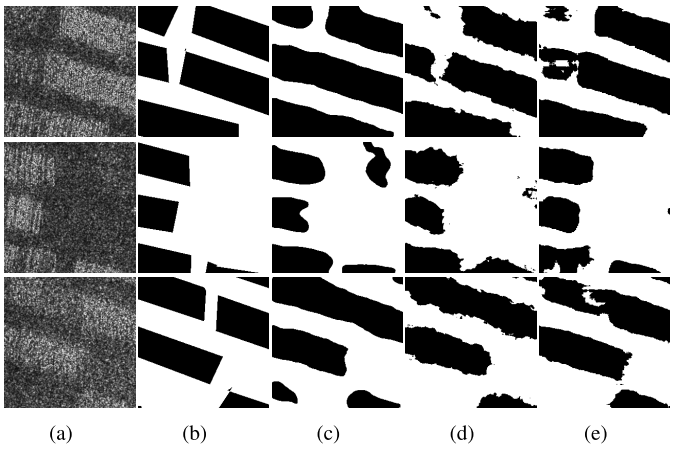


Fig. 5. Experimental results of raft aquaculture based on the supervised deep learning comparison methods. (a) Original image. (b) Ground truth. (c) U-Net. (d) SegNet. (e) IDUDL model.

TABLE I
EVALUATION OF THE COMPARATIVE EXPERIMENTAL RESULTS FOR RAFT AQUACULTURE

Methods	*MIoU*	*Kappa*	*OA*(%)	*P*(%)	*R*(%)
OTSU [42]	0.430 2	0.237 6	61.71	70.54	41.23
K-Means	0.429 6	0.237 1	61.68	70.63	41.00
FCM [43]	0.433 4	0.241 0	61.68	69.96	42.75
SegNet [12]	0.741 4	0.696 9	85.40	84.98	83.40
U-Net [13]	0.746 1	0.699 8	86.01	83.11	86.06
IIC [38]	0.415 1	-	64.67	23.83	61.12
DFCNN [39]	0.440 4	0.183 9	60.02	64.42	45.96
SDL [27]	0.589 3	0.483 6	74.57	85.59	64.28
IDUDL	**0.713 3**	**0.662 4**	**83.16**	**87.10**	**77.52**

between light and dark between the aquaculture area and the sea. However, it does not contain aquaculture semantic information, so it is poor in MIoU and Kappa.

The unsupervised deep learning methods can improve the lack of spatial continuity and noise false positives. As shown in Fig. 4, all the unsupervised deep learning methods reduce the misjudgment of seawater regions due to speckle noise. The IIC uses regular small patches to convert the classical unsupervised point-like into block-like, adding a small spatial continuity. The DFCNN uses the $L1$-norm as a spatial constraint term and is not particularly suitable for SAR aquaculture images with considerable speckle noise. The SDL uses superpixel segmentation to find edge information, but much information

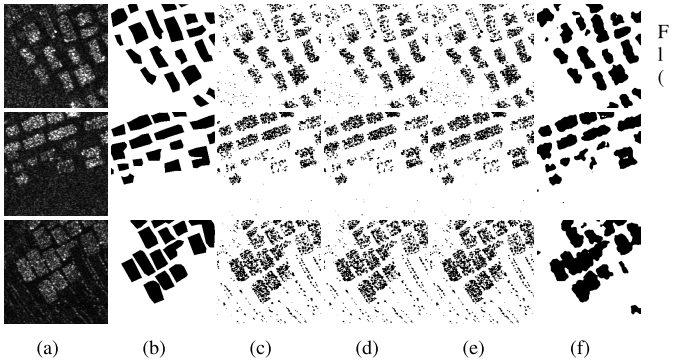


Fig. 6. Experimental results of cage aquaculture based on the classic unsupervised learning comparison methods. (a) Original image. (b) Ground truth. (c) OTSU. (d) K-means. (e) FCM. (f) IDUDL.

is missing in the iteration of a single network, resulting in missed detection in aquaculture areas. They all have a common problem. In the case of completely unsupervised, the network will lose information when overfitting and continue to optimize the problem in the wrong direction. The IDUDL model finds reasonable semantic information for aquaculture and adjusts the optimization direction through double networks.

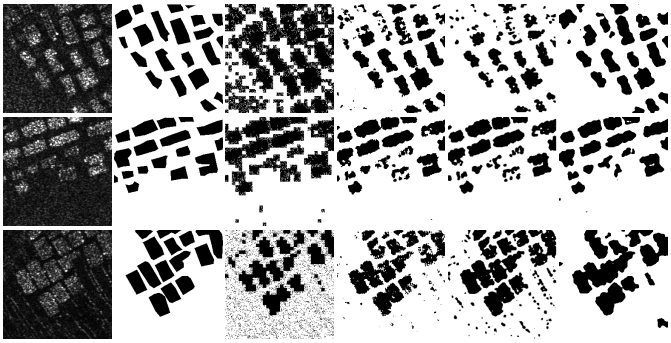
In comparing supervised learning methods, as shown in Fig. 5, SegNet and U-Net showed some "adhesion" for aquaculture, which may be due to upsampling. SegNet only retains position information in the pooling layer. The maximum pooling position restoration will lead to data loss during the upsampling operation. U-Net cuts the feature map of the encoder and expands the decoder feature map. SegNet performs well in the recall, but OA is lower than U-Net. In marine aquaculture, U-Net is more suitable than SegNet, so the IDUDL model chose U-Net as the semantic segmentation network. The experimental results in Table I show that the IDUDL model is more suitable for the aquaculture extraction task than SegNet.

The OA details of the 164 images are shown in Table I. MIoU and Kappa represent spatial consistency, measuring how much semantic information about raft aquaculture is obtained. OA is the classification accuracy of raft aquaculture pixels, which measures the accuracy. P is the extraction correctness of aquaculture information. R is the comprehensiveness of extracting aquaculture. They can measure the degree of influence by coherently speckle noise. While learning the semantics of raft aquaculture, the IDUDL model avoids the interference of speckle noise and improves the OA value. The IDUDL model reduces the missed detection rate, increases R, and rarely extracts nonaquaculture areas as aquaculture, increasing P. The experimental results of the IDUDL model are closest to the supervised learning method. It has achieved good performance in raft aquaculture extraction under completely unsupervised conditions where speckle noise is seriously affected.

D. Cage Aquaculture

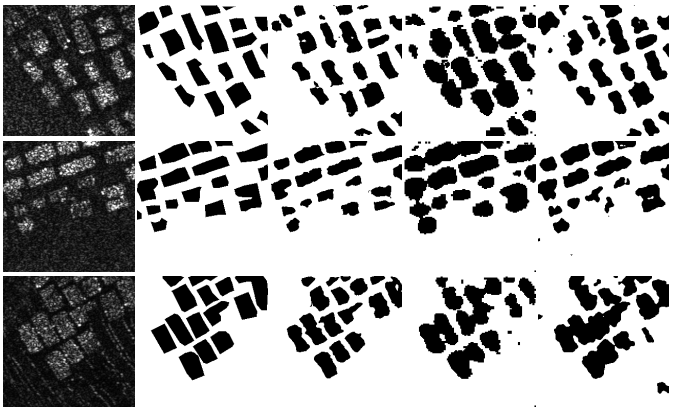
In this section, this article changes the different aquaculture types in different locations under different data. Compared with the raft aquaculture, cage aquaculture is less intensive and, at the same time, affected by different satellite resolutions. There are more extraction targets for cage aquaculture in

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(a) (b) (c) (d) (e) (f)

Fig. 7. Experimental results of cage aquaculture based on the unsupervised deep learning comparison methods. (a) Original image. (b) Ground truth. (c) IIC. (d) SDL. (e) DFCNN. (f) IDUDL.



(a) (b) (c) (d) (e)

Fig. 8. Experimental results of cage aquaculture based on the supervised deep learning comparison methods. (a) Original image. (b) Ground truth. (c) U-Net. (d) SegNet. (e) IDUDL.

one image. Since cage aquaculture is located in a bay sheltered from the wind, and raft aquaculture is situated on the offshore coast, the coherent speckle noise of the cage aquaculture data is more petite.

The classical unsupervised methods show better performance with less speckle noise. OA and *R* of the classical unsupervised methods show that most aquaculture areas are extracted. However, Kappa is poor because of the lack of spatial continuity, as shown in Fig. 6, leading to misjudgment of the sea in the aquaculture area.

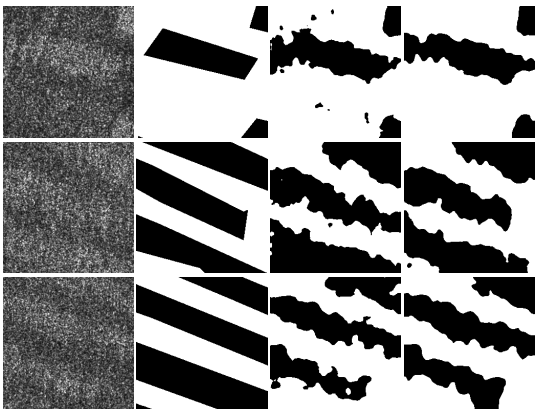
The unsupervised deep learning network for cage aquaculture also has some problems, like raft aquaculture, as shown in Fig. 7. Although IIC improves the continuity of the aquaculture area and has spatial consistency, it still cannot avoid speckle noise in SAR images. In SDL extraction of cage aquaculture, the superpixel algorithm cannot accurately extract the edge because the target is smaller than raft aquaculture. The edge appears jagged, and sea misjudgment occurs in the aquaculture area. DFCNN is affected by the spatial constraints, and there will be black lines on the edge, and raft interference cannot be avoided.

As with the raft aquaculture, the U-Net still showed "adhesion" characters in cage aquaculture in Fig. 8. However, SegNet showed weakened "adhesion," which may be due to smaller and more extraction targets and weakening of coherent

TABLE II

EVALUATION OF THE COMPARATIVE EXPERIMENTAL RESULTS FOR CAGE AQUACULTURE

Methods	*MIoU*	*Kappa*	*OA*(%)	*P*(%)	*R*(%)
OTSU [42]	0.615 8	0.478 0	85.47	87.29	95.06
K-Means	0.615 1	0.476 7	85.46	87.23	**95.14**
FCM [43]	0.615 6	0.478 9	85.10	94.10	87.64
SegNet [12]	0.708 7	0.634 3	88.07	89.99	93.61
U-Net [13]	0.785 1	0.743 0	92.22	97.18	92.82
IIC [38]	0.628 8	0.550 1	78.60	74.75	95.31
DFCNN [39]	0.652 9	0.579 4	84.59	85.05	94.77
SDL [27]	0.714 3	0.640 2	88.78	91.98	92.50
IDUDL	**0.733 9**	**0.673 6**	**89.25**	**94.51**	91.08



(a) (b) (c) (d)

Fig. 9. Incremental experiment results of three regions. (a) Original images. (b) Ground truth. (c) Pretrained IDUDL. (d) Retrained IDUDL.

speckle noise, which resulted in the shrank of the edge of cage aquaculture.

The evaluation of the RADARSAT-2 cage aquaculture data is shown in Table II. The deep learning unsupervised methods find semantic information and improve on both Kappa and MIoU. However, the cage aquaculture semantics in SAR images are not considered, resulting in no significant improvement in OA. With the reduction in speckle noise, the classic unsupervised methods' *R* is significantly improved. IDUDL reasonably enhances the spatial continuity of aquaculture and improves the extraction accuracy while improving the semantic consistency. The final accuracy exceeds the SegNet supervised method in the extraction results of cage aquaculture.

By comparing the two datasets, the IDUDL network improves the common problems of the classical unsupervised methods and unsupervised deep learning methods on the two datasets. It enhances spatial continuity through PE, adds edge information through the SLIC-superpixel algorithm, and further refines semantic information through the U-Net semantic segmentation network. These results are better than other unsupervised methods on two aquaculture extraction datasets and are the closest to the supervised results.

E. Incremental Part

The incremental experiment is based on the raft aquaculture dataset, the convergent semantic segmentation network of the IDUDL model. A total of 106 new images of the same region were input.

The new image is fed into the trained FCSSN to obtain the aquaculture extraction results. These results are called

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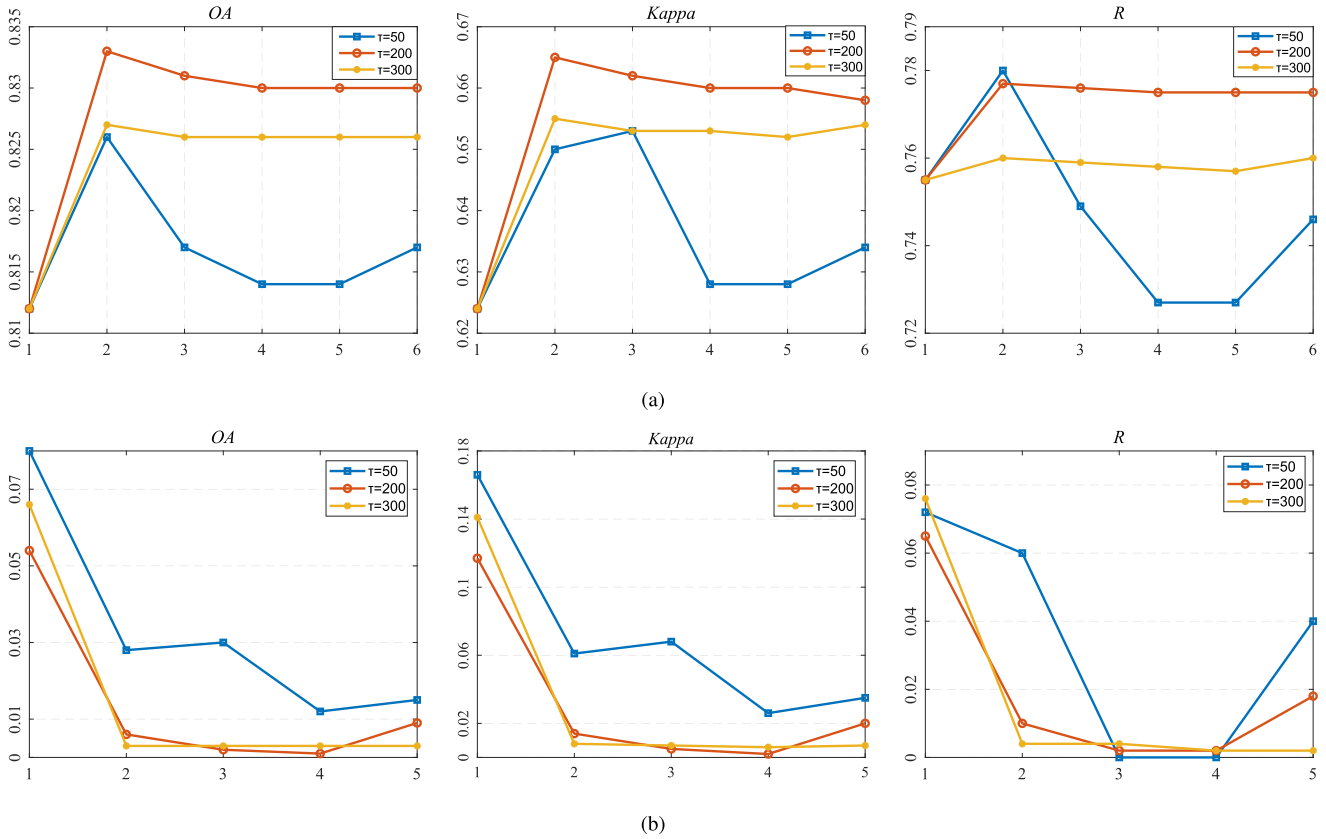


Fig. 10. Experiment of different hyperparameters in pseudo-label accuracy. (a) Difference between pseudo-labels and ground-truth values for each round. (b) Differences in pseudo-labels between every two rounds.

TABLE III
VALUES OF EACH EVALUATION IN INCREMENTAL EXPERIMENT

New_data	*MIoU*	*Kappa*	*OA*(%)	*P*(%)	*R*(%)
Pretrained_results	0.628 8	0.537 1	78.28	77.13	76.09
Retrain_results	0.674 7	0.604 3	81.74	73.07	83.43

the pretrained results. As shown in Table III, different data distributions lead to lower pretrained results accuracy than the original data. As shown in Fig. 9. the pretrained IDUDL model reveals incomplete semantic adaptation to new data. Fig. 9(c) shows serious misjudgment by seawater interference, discontinuous identification of the aquaculture area, and judgment missed. Therefore, incremental training is necessary. FEN uses the pretrained results to generate deep features. PE provides semantically generated pseudo-labels, and the IDUDL model starts incremental training. Fig. 9(d) shows that the retrained IDUDL model adapts to new data semantics with different distributions.

The incremental training results are called the retained results. As shown in Table III, new data consistency and OA are significantly improved after retraining. However, *P* decreased because of retrained results finding more aquaculture areas by semantics. While increasing *R*, *P* will decrease slightly. This is an acceptable increase in inaccuracy. Eventually, retrained results will be close to the results of the original data.

The incremental experiment demonstrates that the IDUDL model's accuracy on new data can be further improved by incremental training, which is significant for the growing amount of unlabeled data.

F. Hyperparameter and Ablation Part

The experiments are carried out on the number of iterations of FCSSN. Too many iterations will make the network converge in advance and fall into local optimal. Too few iterations will make the network unable to converge, and the pseudo-labels keep changing. The experiments are set for 50, 200, and 300 iterations. On one hand, prove that the IDUDL model is updated in the correct direction each round. Comparing the difference between the pseudo-labels and the ground truth in each round evaluates the correctness of IDUDL model learning. On the other hand, determine the convergence condition of the IDUDL model, and set the termination iteration condition by comparing the difference in pseudo-labels between each round.

Fig. 10(a) shows the difference between the pseudo-labels and the ground truth for each round, and Fig. 10(b) shows the difference in the pseudo-labels between every two adjacent rounds. From Fig. 10(a), it can be seen that when $\tau = 50$, the accuracy curve of the IDUDL model fluctuates greatly in each round, resulting in the inability to guarantee the network learning direction. With $\tau = 200$ and $\tau = 300$, the error between the pseudo-labels and the ground-truth in each round conforms to the convergence trend. The gap between the pseudo-labels and the ground truth is closer when $\tau = 200$ than $\tau = 300$. However, to find a suitable termination iteration condition, selecting iteration parameters requires further discussion.

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TABLE IV
RESULTS OF THE ABLATION EXPERIMENTS

Image Enhancement	Patch Estimation	*MIoU*	*OA*(%)
✓		0.542 5	70.79
	✓	0.687 7	81.35
✓	✓	0.713 3	83.16

The IDUDL model judges whether the pseudo-label has converged by comparing the difference between the current pseudo-label and the previous pseudo-label and then whether the network has converged. As shown in Fig. 10(b), among the three evaluation indicators, the stability is the best when $\tau = 300$. However, $\tau = 200$ is closer to the ground truth in the previous conclusion. The difference between pseudo-labels of $\tau = 200$ also has convergence on OA. Therefore, the number of iterations τ of FCSSN is determined to be 200. Correspondingly, the iterative termination condition of the IDUDL model is that when the OA difference δ between the three consecutive rounds of pseudo-labels is less than $\varepsilon = 0.01$, the pseudo-label is considered to be stable.

In addition, FEN also has a hyperparameter κ for the number of iterations. Assimilation often occurs as the number of iterations increases. Coupled with a simple FEN structure, it does not require a lot of iterations to achieve the effect. From the method of PE, the output of FEN is greatly constrained, and the generation of pseudo-labels depends more on superpixel segmentation. Based on the above considerations, the number of iterations $\kappa = 5$ is the optional hyperparameter in this experiment.

Under the same hyperparameter settings, the effects of the image enhancement and PE are discussed by the ablation experiment, as tabulated in Table IV. The semantic information of aquaculture is learned by the PE of IDUDL. So, the accuracy is significantly improved. Image enhancement enables IDUDL to explore information under different views and further improves accuracy.

V. CONCLUSION

This article proposes a novel IDUDL framework to extract the marine aquaculture without any labels in SAR image. With PE and FEN, the proposed method emphasizes the marine aquaculture edge of the pseudo-labels to framework training. It minimizes the pixel similarity impacts of speckle noise. It overcomes the interference of speckle noise in SAR images and introduces marine aquaculture edge semantics information. The semantic segmentation network uses the "adhesion" of U-Net to strengthen further the semantic information of marine aquaculture, which solves the problem of the lack of semantic information in the unsupervised deep learning network. Then, the output of the semantic segmentation network updates the FEN to optimize pseudo-labels. The proposed method can make good use of the newly added data and improve the trained network's generalization, which aims to decrease the training costs. Moreover, it discusses the impact of hyperparameters on the unsupervised networks. The experimental data cover different marine aquaculture methods, indicating the effectiveness of the proposed IDUDL. Extensive experiments on the two SAR datasets and incremental studies demonstrate the advantages of the proposed method in marine aquaculture extraction. In the marine aquaculture extraction task, the proposed method surpasses the classical unsupervised methods. Compared with other unsupervised deep learning, it has more advantages and is closer to the supervised methods. In the future, the aquaculture semantic segmentation model can be extended to handle different types of marine aquaculture synchronously.

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Xinzhe Wang received the Ph.D. degree from the School of Control Science and Engineering, Dalian University of Technology, Dalian, China, in 2012.

He is with the School of Information Science and Engineering, Dalian Polytechnic University, Dalian. His research interests include data-driven modeling and deep learning neural networks.



Jianlin Zhou received the B.S. degree from Dalian Polytechnic University, Dalian, China, in 2020. He is currently pursuing the M.S. degree with the School of Information Science and Engineering, Dalian Polytechnic University.

His main research interests include unsupervised deep learning and remote sensing image processing.



Jianchao Fan (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Control Science and Engineering, Dalian University of Technology, Dalian, China, in 2007 and 2012, respectively.

He was a Post-Doctoral Fellow jointly cultivated by The Chinese University of Hong Kong, Hong Kong, and Dalian University of Technology from 2013 to 2016, and also a Visiting Scholar with the Department of Computer Science and Engineering, Washington University in St. Louis, St. Louis, MO, USA, from 2018 to 2019. He has been a Professor with the Department of Ocean Remote Sensing, National Marine Environmental Monitoring Center, Dalian, since 2019. He has authored/coauthored two books and over 80 scientific articles in international journals and conference proceedings, such as the IEEE TRANSACTIONS ON FUZZY SYSTEMS, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Pattern Recognition*. His research interests include marine SAR image processing, polarimetric SAR (PolSAR), deep learning neural networks, and heuristic swarm intelligent optimization.

Dr. Fan serves as the Program Chair, the Workshop Chair, the Publicity Chair, and the Local Arrangement Chair of several international conferences, such as the International Symposium on Neural Networks, the International Conference on Information Society and Technology, and the International Conference on Intelligent Control and Information Processing.

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